$$\sigma_t = \frac{d_o^2 + d_i^2}{d_o^2 - d_i^2}$$, $d_o = 0.1875$ in (in), $d_o = 0.3125$ in (out)

$$t = \frac{d_i}{2}\left(\sqrt{\frac{\sigma_t + p}{\sigma_t - p}} - 1\right)$$

input: $\sigma_t = (85.13)\frac{(0.1875)^2 + (0.125)^2}{(0.1875)^2 - (0.125)^2}$

$= 221.34$ psi

$$t = \frac{(1.25)}{2}\left(\sqrt{\frac{(221.34) + 85.13}{(221.34) - 85.13}} - 1\right)$$

$t = 0.03125$

output: $\sigma_t = (42.57)\frac{(0.3125)^2 + (0.250)^2}{(0.3125)^2 - (0.250)^2}$

$= 193.93$ psi

$$t = (1.25)\left(\sqrt{\frac{(193.93) + 42.57}{(193.93) - 42.57}} - 1\right)$$

$t = 0.03125$

FBD:

